

[Feature] LRU cache for grammar compiler

#243

New issue

Merged

Ubospica merged 46 commits into `mlc-ai:main` from `DarkSharpness:cache` on Mar 26, 2025

Conversation 48

Commits 46

Checks 38

Files changed 17

+808 -159



DarkSharpness commented

on Mar 16, 2025

Collaborator

No description provided.

Reviewers

Ubospica ✓

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

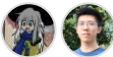
No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants



DarkSharpness added 17 commits 10 months ago

- refactor: decouple the cache and compiler for future update Verified 03045d1
- fix: minor fixes Verified cb051a1
- feat: add a size estimation for grammar type Verified 575f90d
- minor: remove useless comma Verified f8c560b
- feat: add a policy-based thread cache with size limit Verified bf1afa0
- feat: implement an std::list-based LRU map Verified 8adb856
- minor: minor fix Verified ✓ 0cfcede
- feat: try to add a thread safe cache impl Verified 27bceaa
- feat: rewrite in a safe manner Verified ✓ ef8d18d
- feat: complete the thread safe cache Verified 7494609
- feat: integrate the cache into production Verified 48f617a
- minor: add a demo for policy Verified e58da1e

-  [test: add a ctest](#) Verified ✗ [c83faf2](#)
-  [minor: minor fixes](#) Verified ✗ [224b8bf](#)
-  [fix: fix C++ test](#) Verified [36dd49c](#)
-  [chore: update c++ ci | add output and timeout](#) Verified ✗ [ec8b3e9](#)
-  [minor: fix for windows](#) Verified ✓ [c933eb7](#)

 **DarkSharpness** marked this pull request as ready for review [10 months ago](#)

 **DarkSharpness** added 2 commits [10 months ago](#)

-  [feat: rewrite in a simple manner](#) Verified ✓ [804e9d6](#)
-  [fix: fix the estimated size](#) Verified ✓ [358406b](#)



 **Ubospica** reviewed on [Mar 20, 2025](#) [View reviewed changes](#)

cpp/compiled_grammar_data_structure.h Outdated [Show resolved](#)

cpp/support/utils.h Outdated [Show resolved](#)

cpp/support/utils.h Outdated [Show resolved](#)

cpp/support/dynamic_bitset.h Outdated [Show resolved](#)

cpp/grammar_data_structure.h Outdated [Show resolved](#)

 **DarkSharpness** added 6 commits [10 months ago](#)

-  [minor: simplify the interface](#) Verified ✓ [34d21da](#)
-  [minor: minor style improvement](#) Verified [25c7bc0](#)
-  [feat: add python bindings](#) Verified [c04e90c](#)
-  [Merge branch 'main_for_sync' into cache](#) Verified [bfe3e66](#)

- [feat: add python bindings with nanobind](#)

Verified  [3c72b10](#)
- [minor: format](#)

Verified  [5f52bc1](#)



Ubospica reviewed
on [Mar 21, 2025](#)

[View reviewed changes](#)

cpp/grammar_compiler.c

Outdated



Show resolved

cpp/grammar_compiler.c

Outdated



Show resolved

cpp/grammar_compiler.c

Outdated



Show resolved



Ubospica reviewed
on [Mar 21, 2025](#)

[View reviewed changes](#)

include/xgrammar/compiler.h

Outdated



Show resolved

python/xgrammar/compiler.py

Outdated



Show resolved

cpp/support/thread_safe_cache.h

Outdated



Show resolved



Ubospica reviewed
on [Mar 21, 2025](#)

[View reviewed changes](#)

cpp/grammar_compiler.c

Outdated



Show resolved

cpp/grammar_compiler.c

Outdated



Show resolved

cpp/grammar_compiler.c

Outdated



Show resolved

cpp/support/csr_array.h

Outdated



Show resolved

cpp/support/thread_safe_cache.h

Outdated



Show resolved

cpp/grammar_compiler.c
c

Outdated



Show resolved

3 hidden items

[Load more...](#)



DarkSharpness added 5 commits [10 months ago](#)



[style: change the arguments
of constructing the grammar](#)

Verified

[80ec127](#)



[feat: implement std::list
simple ver](#)

Verified

[a74c0ee](#)



[minor: update test](#)

Verified



[6c1fad5](#)



[minor: remove useless import](#)

Verified



[b394faa](#)



[fix: fix test](#)

Verified



[4bcf4c1](#)



Ubospica reviewed
on [Mar 24, 2025](#)

[View reviewed changes](#)

cpp/support/container.
h

Outdated



Show resolved

cpp/support/container.
h

Outdated



Show resolved



DarkSharpness added 6 commits [10 months ago](#)



[test: fix test](#)

Verified

[200a54e](#)



[chore\(c++ ci\): decrease
timeout](#)

Verified

[f522e5d](#)



[minor: doc + rename](#)

Verified

[2b23f37](#)



[perf: optimize performance
when max_size=-1](#)

Verified

[136d2be](#)



[minor: rename list](#)

Verified



[09d48f9](#)



[minor: split list and node
pool](#)

Verified



[c16cb0d](#)



Ubospica reviewed
on Mar 25, 2025

[View reviewed changes](#)

cpp/support/container.
h

Outdated

Show resolved

cpp/support/container.
h

Outdated

Show resolved

python/xgrammar/compiler.py

Show resolved

python/xgrammar/compiler.py

Show resolved

cpp/support/container.
h

Outdated

Show resolved

1 hidden conversation

[Load more...](#)

cpp/support/container.
h

Outdated

Show resolved

cpp/support/container.
h

Outdated

Show resolved

cpp/support/thread_saf
e_cache.h

Outdated

Show resolved

cpp/support/thread_saf
e_cache.h

Outdated

Show resolved

cpp/support/thread_saf
e_cache.h

Outdated

Show resolved



[refactor: simplify code](#)

Verified

[89700fd](#)



Ubospica reviewed
on Mar 25, 2025

[View reviewed changes](#)

cpp/support/thread_saf
e_cache.h

Outdated

Show resolved



DarkSharpness added 4 commits [10 months ago](#)

-   [minor: rename and unify list API](#) Verified [0a1eb3d](#)
-   [feat: add python bindings and python test](#) Verified [b05452b](#)
-   [minor: fix typo](#) Verified ✓ [eec054a](#)
-   [minor: yet another format and fix](#) Verified ✓ [512fe83](#)



Ubospica reviewed
on [Mar 26, 2025](#)

[View reviewed changes](#)

- `cpp/grammar_compiler.cc` Show resolved
- `cpp/grammar_compiler.c` Outdated Show resolved
- `cpp/grammar_compiler.cc` Show resolved
- `python/xgrammar/compiler.py` Outdated Show resolved
- `python/xgrammar/compiler.py` Outdated Show resolved
- `tests/python/test_grammar_compiler.py` Outdated Show resolved
- `tests/python/test_grammar_compiler.py` Show resolved



DarkSharpness added 3 commits [10 months ago](#)

-   [minor: yet another format](#) Verified ✓ [478261d](#)
-   [minor: yet another format](#) Verified ✓ [43de99b](#)
-   [minor: fxxk all trailing return](#) Verified ✓ [c0a9bb2](#)



Ubospica approved these
changes on [Mar 26, 2025](#)

[View reviewed changes](#)

Ubospica left a comment

Collaborator

LGTM. Thanks!



Ubospica merged commit **b18c2c9** into

`mlc-ai:main` on Mar 26, 2025

38 checks passed

[View details](#)



DarkSharpness mentioned this pull request

on Mar 30, 2025

[Minor] Fix missing mark

hf_token_required #284

Merged



Ubospica pushed a commit that referenced this pull request

on Mar 31, 2025



[Minor] Fix missing mark

hf_token_required (#284) ...

Verified



e19752c



russellb added a commit to russellb/vllm that referenced this pull request on Apr 9, 2025



[Core] Upgrade to xgrammar 0.1.18,

add cache size limit ...

549f429



russellb mentioned this pull request on Apr 9, 2025

[Core] Upgrade to xgrammar 0.1.18,

add cache size limit vllm-

project/vllm#16283

Merged



DarkSharpness deleted the `cache` branch

6 months ago